

Customer Health Scoring Model & Automation Spec

SECTION 08-customer-success TYPE document STATUS active OWNER Founder

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Purpose

This document defines the mathematical scoring engine and technical automation spec for Fly GACA Customer Health Scoring. It measures the health of B2B flight schools / Approved Training Organizations (ATOs), charter operators, MROs, and commercial operators licensing Fly GACA Pro for cadet cohorts.

The composite score is a single 0–100 metric calculated automatically every week, driving proactive Customer Success triggers (onboarding, at-risk intervention, renewal, expansion).

It operationalizes [health-dashboard-spec.md](#) and [customer-health-dashboard.xlsx](#).

Composite Formula & Dimension Weights

The composite customer health score (H) is a weighted linear combination of five normalized dimension scores ($S_i \in [0, 100]$):

$$[H = 0.30 \cdot S_{\text{usage}} + 0.20 \cdot S_{\text{engagement}} + 0.20 \cdot S_{\text{support}} + 0.15 \cdot S_{\text{financial}} + 0.15 \cdot S_{\text{sentiment}}]$$

Dimension	Weight ((w _i))	Telemetry Source	Primary Indicator
Product Usage ((S _{usage})))	30%	Firestore progress/summary & cohort telemetry	Seat activation, mock exam volume, Captain Adel AI queries
Engagement ((S _{engagement})))	20%	Web analytics & CS CRM logs	Instructor dashboard logins, session depth, QBR attendance
Support Health ((S _{support})))	20%	Support Ticketing API (Zendesk)	Ticket volume, resolution time, SLA breaches, escalation rate
Financial / Contract ((S _{financial})))	15%	Accounting DB / ZATCA Billing	Seat entitlement utilization, invoice payment timeliness
Sentiment ((S _{sentiment})))	15%	In-app NPS / CSAT & QBR surveys	Cadet NPS, instructor CSAT, executive champion strength

5-Factor Mathematical Calculations

1. Product Usage Score (S_{usage}) — Weight: 30%

Evaluates active participation and learning velocity across the cadet roster over a rolling 7-day window.

$$[S_{\text{usage}} = 0.40 \cdot U_{\text{active}} + 0.30 \cdot U_{\text{exams}} + 0.30 \cdot U_{\text{ai}}]$$

Where:

- **Active Seat Ratio (U_{active}):** $[U_{\text{active}} = \min\left(100, \frac{\text{Active Cadets (7d)}}{\text{Licensed Seats}} \times 100\right)]$
- **Mock Exam Completion Velocity (U_{exams}):** $[U_{\text{exams}} = \min\left(100, \frac{\text{Completed Mock Exams (7d)}}{\text{Licensed Seats}} \times 1.5 \times 100\right)]$
- **AI Instructor Engagement (U_{ai}):** $[U_{\text{ai}} = \min\left(100, \frac{\text{Captain Adel Queries (7d)}}{\text{Licensed Seats}} \times 4.0 \times 100\right)]$

2. Engagement Score ($S_{\text{engagement}}$) — Weight: 20%

Measures organizational alignment, instructor governance, and executive touchpoints over a rolling 30-day window.

$$[S_{\text{engagement}} = 0.40 \cdot E_{\text{instructor}} + 0.30 \cdot E_{\text{depth}} + 0.30 \cdot E_{\text{qbr}}]$$

Where:

- **Instructor Governance ($E_{\text{instructor}}$):** $[E_{\text{instructor}} = \min\left(100, \frac{\text{Instructor Dashboard Logins (30d)}}{\text{Provisioned Instructors}} \times 4.0 \times 100\right)]$
- **Cadet Study Session Depth (E_{depth}):** $[E_{\text{depth}} = \min\left(100, \frac{\text{Average Session Duration (mins)}}{25.0} \times 100\right)]$
- **Executive QBR Participation (E_{qbr}):** $[E_{\text{qbr}} = \begin{cases} 100 & \text{Head of Training / Sponsor attended scheduled QBR} \\ 60 & \text{Designated CFI / Admin attended QBR} \\ 0 & \text{QBR missed, rescheduled >2x, or declined} \end{cases}]$

3. Support Health Score (S_{support}) — Weight: 20%

Quantifies technical friction and customer support load over a rolling 30-day window.

$$[S_{\text{support}} = \max\left(0, 100 - \left(25 \cdot T_{\text{P1}} + 10 \cdot T_{\text{P2}} + 5 \cdot T_{\text{P3}} + 15 \cdot S_{\text{breach}}\right)\right)]$$

Where:

- (T_{P1}): Unresolved P1 Critical tickets (system outage / auth blocking) in last 30d.

- (T_{P2}): Unresolved P2 High tickets (roster sync / score saving error) in last 30d.
- (T_{P3}): Open P3 Medium/Low tickets in last 30d.
- (S_{breach}): First-response or resolution SLA breaches in last 30d.

4. Financial / Contract Health Score ($S_{\text{financial}}$) — Weight: 15%

Tracks contract entitlement adoption and invoice payment compliance.

$$[S_{\text{financial}} = 0.50 \cdot F_{\text{utilization}} + 0.50 \cdot F_{\text{payment}}]$$

Where:

- **Contract Seat Utilization ($F_{\text{utilization}}$):** $[F_{\text{utilization}} = \min\left(100, \frac{\text{Assigned Roster Seats}}{\text{Contracted Seats}} \times 100\right)]$
- **Payment Compliance (F_{payment}):** $[F_{\text{payment}} = \max\left(0, 100 - \max\left(0, \text{Days Past Due} - 14\right) \times 5\right)]$ (100 pts for payment on time or within 14-day grace period; deducts 5 pts per day thereafter).

5. Sentiment Score ($S_{\text{sentiment}}$) — Weight: 15%

Captures subjective customer satisfaction, NPS surveys, and executive relationship stability.

$$[S_{\text{sentiment}} = 0.50 \cdot N_{\text{nps}} + 0.30 \cdot C_{\text{csat}} + 0.20 \cdot K_{\text{champion}}]$$

Where:

- **Net Promoter Score Normalization (N_{nps}):** $[N_{\text{nps}} = \begin{cases} 100 & \text{Promoter (Rating 9–10)} \\ 60 & \text{Passive (Rating 7–8)} \\ 0 & \text{Detractor (Rating 0–6)} \end{cases}]$
- **Customer Satisfaction Rating (C_{csat}):** Average CSAT rating (1–5 scale) normalized: $(C_{\text{csat}} = \text{CSAT Avg} \times 20)$.
- **Executive Champion Index (K_{champion}):** $[K_{\text{champion}} = \begin{cases} 100 & \text{Active executive champion \& public reference logo} \\ 70 & \text{Stable sponsor contact} \\ 30 & \text{Sponsor role changed / champion departed} \end{cases}]$

Score Bands & Recommended CS Playbooks

Band	Composite Score ((H))	Operational Meaning	Recommended Playbook	Automated System Trigger
Healthy	80–100	High adoption, strong ROI proof, zero SLA breaches	Expansion Playbook	Solicit testimonial / case study; notify AE if seat utilization ($\geq 85\%$)
Neutral	60–79	Stable account; minor usage drift or pending QBR	Renewal Playbook	Flag for CSM bi-weekly review; schedule mid-term review

Band	Composite Score ((H))	Operational Meaning	Recommended Playbook	Automated System Trigger
At-Risk	0-59	Churn threat; seat underutilization or support friction	At-Risk Playbook	Immediate P1 CSM alert; initiate 14-day recovery plan

Technical Automation & Data Pipeline Spec

1. Architectural Overview

```

[ Firestore Roster ] ─┐
[ Telemetry Events ] ─┴─> Cloud Scheduler ─> Cloud Function ─> Compute Score ─>
Save Firestore ─> Webhook Alerting
[ Zendesk API ] ─┐ (Weekly Sun 01:00) (calculateHealth) (H composite)
(schools/id/health) (Slack/CRM/Email)
[ ZATCA Invoicing ] ─┐

```

2. Cloud Function Implementation Spec

- **Function Name:** `calculateSchoolHealthScores`
- **Schedule:** Every Sunday at 01:00 UTC+3 (`cron: 0 1 * * 0`).
- **Execution Logic:**
 1. Fetch active school documents from `schools/{schoolId}` .
 2. Aggregate telemetry from `schools/{schoolId}/roster/{cadetUid}` progress summaries over 7d/30d.
 3. Query Support API for ticket counts & SLA breaches in past 30d.
 4. Fetch invoice status from Billing DB (`daysPastDue`).
 5. Compute the 5 dimension scores (S_i) and composite score (H).
 6. Write document to `schools/{schoolId}/health/current` and append snapshot to `schools/{schoolId}/health/snapshots/{YYYY-MM-DD}` .

3. Firestore Document Schema (`schools/{schoolId}/health/current`)

```
{
  "schoolId": "oxford-saudia-dammam",
  "updatedAt": "2026-08-16T01:00:00Z",
  "compositeScore": 84.5,
  "band": "Healthy",
  "dimensions": {
    "usage": 88.0,
    "engagement": 82.5,
    "support": 100.0,
    "financial": 75.0,
    "sentiment": 76.0
  },
  "metrics": {
    "activeSeatRatio": 0.88,
    "weeklyMockExamsPerCadet": 1.8,
    "monthlyInstructorLogins": 12,
    "openTicketsCount": 0,
    "daysPastDue": 0,
    "latestNpsScore": 9
  },
  "previousCompositeScore": 79.0,
  "delta7d": +5.5,
  "triggeredPlaybook": "expansion"
}
```

4. Automated Alert & Escalation Matrix

- **Condition 1 — Score Drops to At-Risk ($H < 60$) or ($\Delta H \leq -15$) pts:**
 - System fires Slack alert to `#cs-at-risk-alerts` with detailed dimension breakdown.
 - Automatically creates Urgent P1 Task in Hubspot/Salesforce CRM assigned to CSM.
 - Triggers [At-Risk Playbook](#) workflow.
- **Condition 2 — Seat Utilization ($\geq 85\%$) & ($H \geq 80$):**
 - System fires Slack alert to `#sales-expansion`.
 - Sends email to Account Executive with upsell recommendation (e.g., +25 seat expansion proposal).
- **Condition 3 — Support Health ($S_{\text{support}} < 50$):**
 - Sends automated high-priority ticket escalation to Head of Product & CS Lead.

Related Documents

- [Health Dashboard Spec](#)
- [At-Risk Playbook](#)
- [Renewal Playbook](#)

- [Expansion Playbook](#)
- [Customer Success Overview](#)